

TELCONG NETPLAN

Network Planning & Regulatory Affairs

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For incorporation into Section D-1 of Form NCC-7B

3GPP Release 18 Features

Implementation notes for the 5G CBD Pilot technical filing

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Scope	5G NR features standardised in 3GPP Release 18 (a.k.a. 5G-Advanced)
Use	Technical justification depth, Form NCC-7B Section D
Source	3GPP TSG-RAN plenary meeting outputs, RP-241200 series

Background

3GPP Release 18 (sometimes branded 5G-Advanced) was frozen in June 2024 with ASN.1 freeze in September 2024. It is the first 5G release to incorporate native AI/ML support into the air interface, alongside significant enhancements to reduced-capacity (RedCap) devices, non-terrestrial network (NTN) integration, sidelink, and energy-efficiency features. For the Abuja CBD 5G pilot, Release 18 features are relevant because the Tachyon equipment we are deploying (5G NR Massive MIMO radio unit 64TR ed.2 and BBU-7600 baseband processor) supports Release 18 features in firmware, even though the initial deployment will operate primarily in Release 16/17 modes.

Section 1, Reduced-Capacity NR (RedCap) enhancements

Release 18 introduces eRedCap, an enhanced reduced-capacity device profile targeting wearables, video surveillance, and low-end IoT. The eRedCap profile reduces device complexity further than Release 17 RedCap by limiting the maximum bandwidth to 5 MHz (down from 20 MHz in Release 17), reducing the receive antenna count, and simplifying the MIMO requirements.

Implications for the Abuja CBD pilot: the Tachyon equipment supports eRedCap in firmware. While the initial pilot deployment will not target eRedCap devices, the equipment is forward-compatible. We should note this in Section D as evidence of equipment longevity and forward-compatibility.

Section 2, Sidelink enhancements (NR sidelink Phase 3)

Release 18 advances NR sidelink (device-to-device) into its third phase, with support for the unlicensed band (NR sidelink-U), positioning enhancements, and relay improvements. NR sidelink is primarily relevant for V2X (vehicle-to-everything) and public safety applications. For an urban densification pilot in Abuja CBD, sidelink is not a primary use case, but the technology is mentioned here for completeness and because the Commission has asked about it in past hearings.

Implications for the Abuja CBD pilot: low relevance. Mention in Section D as a feature the Tachyon equipment supports, but do not commit to V2X or public safety deployment in the pilot scope.

Section 3, AI/ML for the air interface

Release 18 is the first release to standardise AI/ML support in the 5G NR air interface. The standardisation covers three use cases: (a) beam management, (b) channel state information (CSI) feedback enhancement, and (c) positioning accuracy improvement. The framework allows the model training to occur either at the network side, the device side, or jointly, with model lifecycle management (deployment, monitoring, update, retirement) coordinated across the air interface.

Implications for the Abuja CBD pilot: the Tachyon equipment supports the network-side AI/ML beam management use case in firmware. Initial deployment will use the deterministic Release 16 beam management algorithms; we will evaluate enabling AI/ML beam management in the post-pilot phase, contingent on validated improvements in beam alignment KPIs. This is forward-looking and should be framed carefully in Section D, the Commission will ask about training data provenance and model bias.

Section 4, Non-Terrestrial Network (NTN) evolution

Release 18 advances NTN integration with full LEO satellite support, regenerative payload architectures, and enhanced mobility management for satellite handover. NTN is highly relevant for rural coverage but not for the urban Abuja CBD pilot. The feature is mentioned because the Commission has indicated interest in NTN as a complement to terrestrial rural coverage in subsequent regulatory cycles.

Section 5, Energy efficiency features

Release 18 introduces a set of energy-efficiency features including network-energy-savings (NES) modes, bandwidth part adaptation, and traffic-aware cell on/off cycling. These features can reduce RAN energy consumption by 20-30% in low-traffic periods, with no impact on user experience during high-traffic periods. For Abuja CBD, where load varies significantly between business hours and overnight, NES is highly relevant.

Implications for the Abuja CBD pilot: the Tachyon equipment supports NES Mode 2 (bandwidth part adaptation) and NES Mode 4 (slot-level on/off). We commit in Section D to enabling NES Mode 2 at pilot launch and evaluating Mode 4 enablement within the first 12 months. This commitment supports the local content / environmental impact narrative.

Section 6, Summary table

Release 18 feature	Relevance to Abuja CBD pilot	Section D treatment
eRedCap	Low (initial scope)	Mention as forward-compatibility
NR Sidelink Phase 3	Low (urban use case)	Brief mention
AI/ML beam management	Medium (post-pilot)	Forward-looking commitment
AI/ML CSI feedback	Low (initial scope)	Mention only
AI/ML positioning	Low (initial scope)	Mention only
NTN integration	Low (urban use case)	Mention only, rural relevance noted

Release 18 feature	Relevance to Abuja CBD pilot	Section D treatment
NES Mode 2 (bandwidth part adaptation)	High	Commitment at launch
NES Mode 4 (slot on/off)	Medium (12-month target)	Forward commitment
Network slicing enhancements	Medium (enterprise services)	Mention as enterprise capability
URLLC enhancements	Low (initial scope)	Mention only

END OF NOTES

For incorporation into Section D-1 of Form NCC-7B. Cross-reference 3GPP RP-241200 series for source plenary documents.